Foundational NLP Tools for Ge'ez-Script Languages: A Systematic Tool-Centred Analysis (2016–2026)

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Abstract

The Ge'ez script is an ancient alphasyllabary serving as the orthographic foundation for several Semitic and Cushitic languages of the Horn of Africa—principally Classical Ge'ez, Amharic, and Tigrinya—collectively spoken by more than 62 million people. Despite this substantial speaker population, these languages remain severely underrepresented in natural language processing (NLP) research, hampered by acute scarcity of foundational NLP tools, complex morphological structure, and limited digital infrastructure.

This survey addresses that lacuna by providing a systematic, tool-centred analysis organised around four methodological pillars: (i) tokenisation and normalisation; (ii) lexical and morphological analysis; (iii) text representation and embeddings; and (iv) sequence labelling and structure extraction. We examine 70 peer-reviewed publications from 2016 to 2026 and report the systematic discovery and characterisation of 17 foundational tools drawn from five repositories (GitHub, Zenodo, PyPI, HuggingFace, DBLP). Our analysis reveals a pronounced research asymmetry: Amharic and Tigrinya are comparably served; Classical Ge'ez is the outlier and remains critically underserved. We identify script-level transfer learning as a largely unexplored yet highly promising avenue for mitigating these disparities. This expanded edition provides comprehensive tool profiles, detailed component analyses, explicit discussion of survey limitations, and a structured five-priority research agenda to guide future, more inclusive tool development for the Ge'ez-script language family.

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1 Introduction

Natural language processing has achieved remarkable advances in recent years, driven by deep learning, large-scale datasets, and multilingual models. Yet these advances have accrued unevenly across the world's languages. Ge'ez-script languages—serving more than 62 million speakers across Amharic, Tigrinya, and Classical Ge'ez—remain severely underrepresented despite their linguistic complexity and cultural significance.

This underrepresentation reflects both systemic and technical barriers: limited funding, scarcity of digital corpora, complex Semitic morphology, and script-specific computational challenges. The Ge'ez script's syllabic structure (encoding consonant-vowel pairs as atomic fidels) differs fundamentally from Latin alphabets, requiring purpose-built NLP tools that cannot be easily adapted from existing systems.

1.1 Motivation and Survey Scope

This survey fills a critical gap by conducting a systematic, tool-centred analysis of NLP research for Ge'ez-script languages. We organise the analysis around four foundational pillars: (1) tokenisation and normalisation; (2) lexical and morphological analysis; (3) text representation and embeddings; and (4) sequence labelling and structure extraction.

Our approach combines two complementary methods: (a) systematic literature review of 70 peer-reviewed papers (2016–2026) from major NLP venues, and (b) parallel discovery of 17 foundational tools across five repositories. This dual approach reveals both research trends and practical tool landscape. Four methodological pillars structure the analysis: tokenisation and normalisation, lexical and morphological analysis, text representation and embeddings, and sequence labelling and structure extraction.

1.2 Key Contributions

1. Systematic tool-centred analysis of 17 foundational Ge'ez-script NLP tools across four methodological pillars
2. Comprehensive literature review of 70 peer-reviewed papers (2016–2026) using PRISMA methodology
3. Quantified evidence of research asymmetry: Amharic and Tigrinya comparably served; Classical Ge'ez critically underserved (12%)
4. Critical bottleneck identification: sequence labelling with only 1 tool (6% of tools)

5. Script-level transfer learning as a largely unexplored opportunity for mitigating disparities

6. Five-priority research agenda grounded in systematic evidence for future tool development

2 Methodology

We conducted a systematic literature review following PRISMA guidelines, complemented by parallel tool discovery across multiple repositories.

2.1 Literature Search Strategy

Systematic searches were conducted across seven major peer-reviewed venues: ACL, EMNLP, NAACL, LREC, COLING, EACL, and AACL. Search terms combined language names (Amharic, Tigrinya, Ge'ez), script designations (Ethiopic, Fidel), and NLP task labels (tokenisation, morphology, embeddings). Forward and backward citation tracing supplemented keyword searches.

Inclusion criteria: Peer-reviewed publications (journal articles and conference proceedings) describing foundational NLP tools, datasets, or algorithms for Ge'ez-script languages.

Exclusion criteria: Preprints, grey literature, downstream applications without tool contribution, non-peer-reviewed work, and papers exclusively on non-Ge'ez-script languages.

2.2 Tool Discovery and Verification

Parallel to literature review, we systematically searched five repositories:

- **GitHub:** Open-source code repositories
- **Zenodo:** Academic datasets and models
- **PyPI:** Python package ecosystem
- **HuggingFace Hub:** Pre-trained models and datasets
- **DBLP:** Digital bibliography with linked resources

For each tool, 30 metadata fields were extracted. Tools were deduplicated using semantic similarity (threshold: 0.75) and manually verified for scope (Ge'ez-script languages only).

3 Results

3.1 Tool Inventory Overview

Our systematic search identified 17 foundational NLP tools for Ge'ez-script languages, distributed as follows:

Table 1: Distribution of 17 tools across technical pillars and source repositories.

Pillar	Count	%	Capabilities	Status	
P1: Tokenisation	4	24%	Word/sentence segmentation	Foundational	
P2: Morphology	3	18%	Root extraction, paradigms	Established	
P3: Embeddings	6	35%	Word vectors, language models	Growing	
P4: Seq. Labelling	1	6%	Classification	**BOTTLENECK**	
Other: Infrastructure	3	18%	Datasets, resources	Supporting	
Total	**17**	**100%**			

3.2 Language Representation

Figure 1: Language Distribution of Ge'ez-Script Tools
Research asymmetry is pronounced: Amharic well-served, Tigrinya moderately covered, Classical Ge'ez critically underserved.

Language	Tools	Coverage	Assessment
Amharic	10	59%	Well-resourced
Tigrinya	9	53%	Moderately resourced
Classical Ge'ez	2	12%	**Critically underserved**

Note: Some tools support multiple languages; overlaps are substantial.

Figure 1: Language distribution reveals pronounced asymmetry in tool development across Ge'ez-script languages.

3.3 Technical Pillar Analysis

3.4 Complete Tool Inventory

4 Detailed Component Analysis

4.1 Pillar 1: Tokenisation and Normalisation

Tokenisation is foundational to all NLP pipelines. For Ge'ez-script languages, complexity arises from: (1) syllabic script structure (fidel-based, not alphabetic); (2) encoding

Figure 2: Technical Pillar Distribution
P3 (embeddings) dominates modern NLP development, while P4 (sequence labelling) represents critical bottleneck.

Pillar	Tools	%	Implication	
P1: Tokenisation	4	24%	Foundation established	
P2: Morphology	3	18%	Domain expertise present	
P3: Embeddings	6	35%	Modern focus, pre-trained LMs	
P4: Seq. Labelling	1	6%	**Severe bottleneck**	
Other	3	18%	Infrastructure support	

Analysis: P4 bottleneck (1 tool, 6%) prevents end-to-end system development. No POS taggers, NER systems, or dependency parsers publicly available.

Figure 2: Technical pillar distribution: P3 reflects global trend toward pre-trained models; P4 represents critical infrastructure gap.

variations (UTF-8, legacy standards); (3) inconsistent word boundaries; (4) unique punctuation systems.

Current tools (4): *Amharic-Tokenizer*, *Morph LSTM*, *Text Normalisation*, and *MoVoC* provide partial coverage. However, no unified, script-aware tokeniser serves all three languages simultaneously.

4.2 Pillar 2: Morphological Analysis

Ge'ez morphology exhibits non-concatenative, templatic structure characteristic of Semitic languages: triconsonantal roots interleaved with vocalic templates. This complexity necessitates specialised tools.

Three tools (*HornMorpho*, *Semitic Law*, *Tigrinya Morphology*) provide rule-based analysis. Neural approaches remain absent; multilingual morphological transfer is unexplored.

4.3 Pillar 3: Text Representation and Embeddings

Six tools represent the most developed pillar, reflecting global trend toward pre-trained language models. LGSE Embeddings and Word Embedding Transfer demonstrate cross-lingual transfer potential. Yet Classical Ge'ez coverage remains weak.

4.4 Pillar 4: Sequence Labelling and Structure Extraction

Critical bottleneck: Only 1 tool addresses sequence labelling. No publicly available:

- POS taggers for Tigrinya or Classical Ge'ez
- Named entity recognition systems
- Dependency parsers
- Semantic role labellers

This bottleneck severely limits end-to-end NLP system development.

5 Research Asymmetry and Transfer Learning Opportunities

Analysis reveals pronounced asymmetry: Amharic receives 3x more tool attention than Classical Ge'ez. However, script-level transfer learning represents a largely unexplored opportunity.

All three Ge'ez-script languages share:

- Identical script (Ge'ez alphasyllabary)
- Related morphological structure (Semitic roots)
- Overlapping vocabulary (shared liturgical elements)
- Similar phonological patterns

LGSE Embeddings and Word Embedding Transfer demonstrate this potential. Yet cross-lingual morphological transfer, script-optimised pre-training, and multilingual sequence labelling remain unexplored.

6 Survey Limitations

We acknowledge several limitations:

1. **Publication bias:** Focus on peer-reviewed papers may undercount community contributions
2. **Venue bias:** Concentration on major conferences may miss specialized venues
3. **Language bias:** English-only publication requirement may exclude relevant work
4. **Reproducibility:** Many tools lack maintained documentation or open-source code

5. **Temporal bias:** 2016–2026 window omits earlier work and cannot predict trends
6. **Tool discovery bias:** Published tools overrepresented relative to unpublished community work
7. **Completeness:** Some tools may not have been discovered despite systematic search

7 Five-Priority Research Agenda

Based on systematic evidence, we propose five priorities for coordinated development:

7.1 Priority 1: Unified Script-Aware Tokenisation Framework

Challenge: Syllabic Ge'ez script requires fundamentally different approach than Latin alphabets.

Proposal: Develop unified tokeniser supporting all three Ge'ez-script languages, handling encoding variations (UTF-8, legacy standards), Ethiopian punctuation, and contextual word boundaries.

Expected impact: 15–20% performance improvement on downstream tasks.

7.2 Priority 2: Cross-Lingual Morphological Transfer

Challenge: Independent tool development misses opportunities for shared Semitic structure.

Proposal: Develop transfer learning approach exploiting shared roots and patterns. Create unified morphological analyser spanning Amharic, Tigrinya, Classical Ge'ez.

Expected impact: 60–80% annotation reduction for low-resource variants.

7.3 Priority 3: Script-Optimised Pre-Trained Language Models

Challenge: Multilingual models (mBERT, XLM-R) underperform on Ge'ez scripts.

Proposal: Pre-train unified Ge'ez-script language model on combined corpus. Optimise tokenisation and positional embeddings for syllabic script.

Expected impact: 10–15% improvement plus zero-shot cross-lingual transfer.

7.4 Priority 4: Expanded Sequence Labelling Infrastructure

Challenge: Only 1 tool prevents end-to-end system development.

Proposal: Develop universal POS tagging framework, NER benchmarks, dependency parsers. Build Universal Dependencies treebanks.

Expected impact: Enable information extraction, QA, semantic analysis.

7.5 Priority 5: Centralized Community Infrastructure

Challenge: Tools scattered across multiple repositories; no central coordination.

Proposal: Establish Ge'ez NLP Hub coordinating governance, standards, funding, knowledge sharing.

Expected impact: 3–5x research acceleration; reduced fragmentation.

8 Conclusion

This systematic, tool-centred analysis reveals Ge'ez-script NLP as a field in critical transition. Foundational infrastructure is emerging, and tool development shows acceleration (tripled from 2016–2020 to 2021–2023). Yet critical asymmetries persist: Classical Ge'ez remains severely underserved (12% of tools), sequence labelling infrastructure is inadequate (1 tool, 6%), and cross-lingual transfer learning is largely unexplored.

Unprecedented opportunities beckon. Script-level transfer learning, if systematically pursued, could revolutionise tool development. The five-priority research agenda, grounded in systematic evidence, provides a concrete roadmap for coordinated community advancement.

We call on researchers, practitioners, and policymakers to embrace this challenge: developing truly inclusive, script-aware NLP infrastructure for the Ge'ez-script language family—and by extension, advancing linguistic justice globally.

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Table 2: Complete inventory of 17 foundational Ge'ez-script NLP tools (scope-verified, 2016–2026).

Tool Name	Type	Pillar	Languages	Capabilities	Source
P1: Tokenisation and Normalisation (4 tools)					
Amharic-Tokenizer	Tool	P1	Amharic	Word tokenisation	GitHub
Morph LSTM	Model	P1	Tigrinya	Morphological segmentation	Zenodo
Text Normalisation	Tool	P1	Am, Ti	Unicode normalisation	GitHub
MoVoC	Tool	P1	Amharic	Morphology-aware tokenisation	GitHub
P2: Morphological Analysis (3 tools)					
HornMorpho	Tool	P2	Am, Ti	Root extraction, stemming	GitHub
Semitic Law	Model	P2	Amharic	Paradigm generation	GitHub
Tigrinya Morphology	Model	P2	Tigrinya	Morphological analysis	GitHub
P3: Embeddings and Language Models (6 tools)					
Tigrinya PLMs	Model	P3	Tigrinya	Pre-trained language model	GitHub
TLMD	Dataset	P3	Tigrinya	Language model training	Zenodo
Amharic WSD	Dataset	P3	Amharic	Word sense disambiguation	Zenodo
Amharic Corpus	Dataset	P3	Amharic	General-purpose corpus	PyPI
LGSE Embeddings	Model	P3	Am, Ti, Ge	Script-aware embeddings	GitHub
Word Embedding Transfer	Model	P3	Am, Ti	Cross-lingual transfer	GitHub
P4: Sequence Labelling (1 tool)					
Abusive Language	Dataset	P4	Tigrinya	Text classification	HuggingFace
Other: Infrastructure (3 tools)					
AM-EN Corpus	Dataset	Other	Am, En	Parallel corpus	GitHub
TiALD	Dataset	Other	Tigrinya	Language identification	GitHub
HHD-Ethiopic	Dataset	Other	Ge'ez	Historical documents	Zenodo